

AUTHENTICATED INGRESS BOUNDARY

A Hardware-Enforced Secure Ingress Barrier for Manchester/RF
Serial Links

Category	IC Chip Design & FPGA Implementation
Focus Area	04 - Secure Communication (secure framing & interface integrity)
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Source	docs/proposal/salaras-rx-proposal.en.md

1. Executive Summary

Problem:

Lightweight serial/RF receivers parse an untrusted bitstream into a shift register and expose an integrity field to the host without validating it.

This gap is measured on the Tiny Tapeout 07 Manchester baseline `tt07-bep-decode`: the 24-bit integrity field (`tail_1..3`) is received but never checked, so a corrupt payload or a fault-injected frame still appears valid to the host (CWE-354).

Solution:

A fail-closed, reusable ingress boundary: a keyed MAC (SIMON-32/64, fixed-length CBC-MAC) plus a freshness counter and an atomic commit.

It closes forgery (CWE-345), replay (CWE-294), control/data de-synchronization (CWE-1264), invalid framing (CWE-20), and a hung FSM (CWE-1245).

Chip:

A small pure-digital IP with a single clock in the committed scope, targeting sky130 ASIC and the DE10-Nano FPGA.

Users:

Secure-element and contactless smart-card designers, IP integrators needing an ingress-hardening block, and firmware teams that need assurance that a frame read from the host is authenticated.

Measured results:

- 128 of 128 single-bit flips rejected.
- Forgery, wrong key, replay, and stale counter rejected; 20 clean frames accepted (false reject 0).
- End-to-end latency 108 cycles (107 for MAC and freshness, 1 for commit).
- Five formal properties pass, proving the fail-closed structure.
- RF appendix: real sky130 hardening on a 1x2 tile, die 0.0363 mm², WNS 0.00, typical power 1.21 mW.

Impact:

A corrupt, forged, or replayed frame no longer appears valid to the host, with measurable and traceable assurance.

2. Background & Problem Statement

Background:

Serial and RF links carry data between transceivers, terminals, and hosts in identity and payment systems.

In lightweight designs the physical line coding (Manchester, 8b/10b) is treated as sufficient, so the receiver parses an untrusted bitstream and no frame is authenticated.

Two failure classes follow.

First, no validation of the received integrity value (CWE-354): a keyless CRC or ECC detects random errors but is forgeable by an active attacker, and an unimplemented check provides nothing.

Second, non-atomic commit (CWE-1264): data and control can de-synchronize under glitches or jitter.

A fragile FSM (CWE-1245) and unvalidated input (CWE-20) compound it.

Evidence from a real link:

The baseline `tt07-bep-decode` (433 MHz Manchester, Tiny Tapeout 07) receives a 24-bit integrity field and exposes it to the host without checking it.

We reproduced the consequence in simulation: the baseline latches a corrupted payload and a corrupted integrity field with `full=1` (CWE-354, measured).

The field is not a standard CRC but an undocumented error-correcting code; this is confirmed by the baseline author, and the algorithm is unsolved.

The RF case is therefore evidence of the vulnerability class, not our solved integrity path.

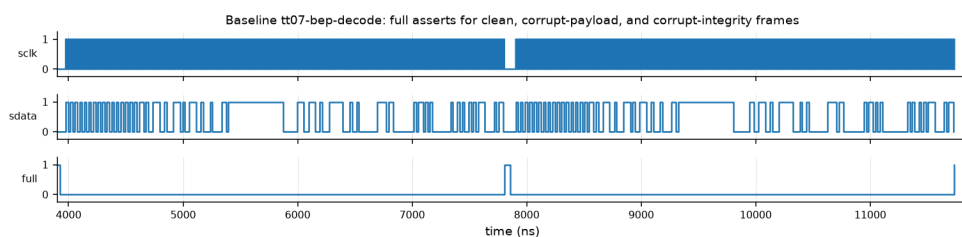


Figure: `full` stays high for a clean frame, a corrupt payload, and a corrupt integrity field (baseline `tt07-bep-decode`).

Gap vs available solutions:

Approach	Limit
Parity or CRC byte	Detects random errors; no framing; forgeable
Full 8b/10b	DC balance only; no integrity

Approach	Limit
Software verification	Runs after data crosses the trust boundary
Standard decoders	Validate only the start of the frame
MAC without freshness	Forge-resistant but replayable

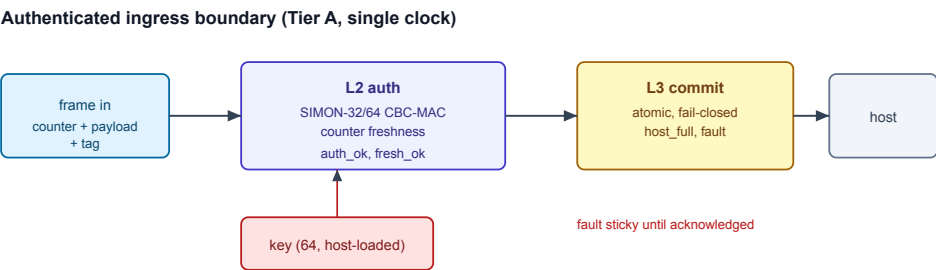
None combine authentication, freshness, and atomic fail-closed commit at the boundary in a small area.

Problem statement:

1. How to authenticate a received frame against forgery (CWE-345)?
2. How to reject replayed frames without persistent state (CWE-294)?
3. How to commit data and control atomically, fail-closed, so a failed check never reaches the host (CWE-1264, CWE-1245)?
4. How to make this a reusable core that fits different front-ends (Manchester/RF, SerDes, UART) and both FPGA and ASIC?

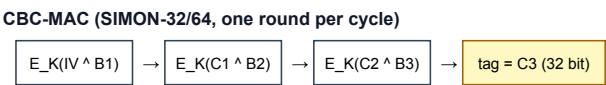
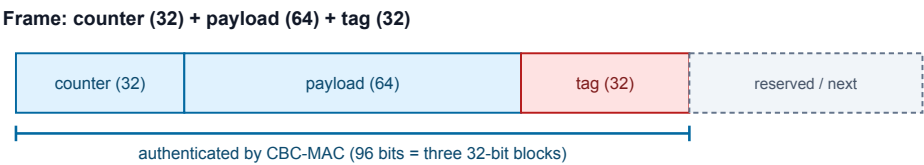
3. Proposed Chip Design

3.1 System Architecture



No buffer, no block RAM. host_full is asserted only when auth_ok and fresh_ok are both high at frame end.

Figure: authenticated boundary architecture (Tier A, single clock).



IV = 0, 64-bit key, 32-bit tag. The tag is compared against the received tag; the counter must be strictly greater than the last accepted.

Figure: link frame format and the CBC-MAC chain.

Modules:

- `simon32_64.v`: serialized SIMON-32/64 block cipher, one round per cycle.
- `l2_auth.v`: CBC-MAC over counter plus payload (three 32-bit blocks), tag compare, and counter freshness.
- `l3_commit_gatekeeper.v`: atomic commit only when authentication and freshness pass; on failure it holds `host_full` low and raises a sticky `fault`.
- `salaras_auth_top.v`: L2 and L3 integration.
- `project.v`: Tiny Tapeout wrapper with a serial frame loader.

Interface:

- Inputs: 64-bit key, 32-bit counter, 64-bit payload, 32-bit tag.
- Outputs: `host_full`, `host_data`, `fault`, `auth_ok`, `fresh_ok`, `done`.
- `full` handshake and `fault` notification; `fault` is sticky until acknowledged.

Processing and memory:

- Streaming data path; CBC-MAC runs as blocks arrive.
- No block RAM; state is registers only (MAC, counter, flags).

Power:

- Single-clock digital logic, no DSP, internal PLL, or large memory.
- The MAC is active only during a frame, so switching activity is minimal when idle.

3.2 Resource Estimate

FPGA estimate (Yosys, before Quartus synthesis):

Resource	Estimate	DE10-Nano capacity
Logic elements / LUT	about 360 LUT equivalent	41,910 ALM
Registers / flip-flops	500	415,000
Block RAM (M10K)	0	5,570 Kbit
DSP	0	112

ASIC target: Tiny Tapeout sky130 130nm via OpenLane.

The RF appendix has a real result: 1x2 tile, die 0.0363 mm², WNS 0.00, typical power 1.21 mW.

Sky130 hardening for the new link: 2x2 tile, die 0.0756 mm², 2354 cells, 0 DRC, 0 LVS, WNS 0.00, typical power 1.87 mW.

Tools:

- Intel Quartus Prime (synthesis, fit, timing, SignalTap) for the DE10-Nano.

- OpenLane and Yosys for the sky130 ASIC path.
- Verilator and Icarus Verilog for simulation and lint.
- cocotb and pytest for automated testbenches.
- SymbiYosys for formal proofs.

3.3 Test Plan

RTL simulation (S1, measured):

- cocotb drives the MAC, L2, and commit with the success matrix: clean frames accepted, forgery, wrong key, replay, and stale counter rejected, and 128 of 128 single-bit flips rejected.
- Latency measured per stage: 33 cycles per SIMON block, 107 cycles for MAC and freshness, 108 cycles end to end.

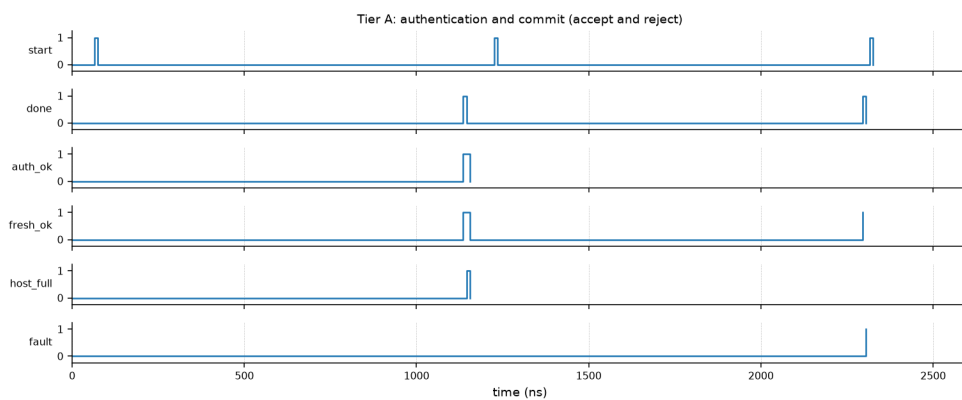


Figure: a clean frame accepted and a corrupted frame rejected at the authenticated boundary.

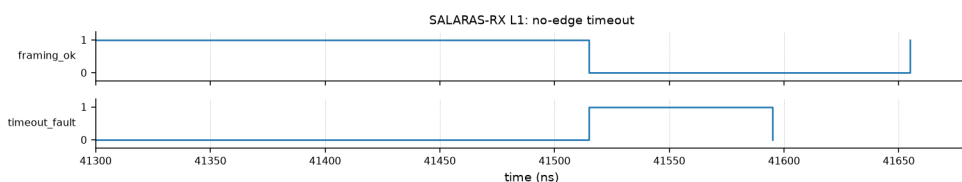


Figure: a timeout fault on the boundary path (RF appendix evidence).

DE10-Nano board test (S2, planned):

- Synthesis procedure: a Quartus project with a board wrapper, SDC, and pin assignments.
- Bitstream implementation (.sof/.rbf) and real-time on-board testing.
- Internal signal verification with SignalTap on `auth_ok`, `fresh_ok`, `done`, `host_full`, and `fault`.
- On-board test shows a clean frame accepted and a corrupt frame rejected.
- The organizer's FPGA/sandbox facility is used at the bootcamp to run this stage.

Success metrics:

Metric	Target	Evidence
Single-bit error detection	100 percent	Simulation (128/128)
False reject	0 percent	Simulation (20 clean frames)
Forgery and replay	rejected	Simulation
End-to-end latency	measured 108 cycles	Simulation
Fail-closed	formally proven	SymbiYosys

4. References

- PERURI. "Buku Panduan Peserta PERURI Chip Hackathon 2026." <https://summit.peruri.co.id/docs/Buku-Panduan-PERURI-Chip-Hackathon.pdf>
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- Kohnen, Z. and Alvarado, A. "Manchester decoder of a home thermostat's wireless protocol." FSiC, 2025.
- Beaulieu, R. et al. "The SIMON and SPECK Families of Lightweight Block Ciphers." IACR ePrint 2013/404.
- Tiny Tapeout. <https://tinytapeout.com/>
- MITRE. CWE-354, CWE-345, CWE-294, CWE-1264, CWE-1245, CWE-20.
- Terasic. "DE10-Nano - Cyclone V FPGA Guide."

5. Appendix

Appendix A. Team & Roles

Name	Expertise	Role
Ibrahim Fauzi Rahman	RTL / Verilog	RTL designer, integration, synthesis
Idris Syaifulloh	Verification / Python	cocotb, fault injection, metrics
Dr. Setia Jul Ismail, S.T., M.T.	Architecture / Methodology	Advisor, claim validation

Appendix B. Outputs & Demo

- Verilog RTL, cocotb testbench scripts, and formal properties.
- FPGA bitstream (.sof/.rbf) and on-board demo (bootcamp plan).
- Source repository and a short technical report.

Appendix C. Bootcamp Plan (18-20 October 2026)

Day	Focus	Deliverable
Day 1	Finalize link RTL, SerDes integration, start advanced crypto analysis	Simulatable RTL
Day 2	Quartus synthesis, SignalTap, on-board fault-injection and replay tests	Bitstream and resource/timing report
Day 3	Final measurements, hardening polish, demo, Top 5 to Top 3 presentation	Demo and presentation material

Appendix D. Latency Budget (measured)

Stage	Value
SIMON block	33 cycles
MAC and freshness	107 cycles
Commit	1 cycle
End to end	108 cycles

Appendix E. Integrity Comparison (measured)

Property	Keyless CRC	Keyed MAC	MAC + counter
Latency	73 cycles	107 cycles	108 cycles
Random error detection	yes	yes	yes
Forgery resistance	no	yes	yes
Replay resistance	no	no	yes

Appendix F. Problem Evidence (RF)

The baseline `tt07-bep-decode` latches a corrupt payload and integrity field with `full=1` (CWE-354, measured). Details in `sim/RESULTS.md`.

Appendix G. Limits

- No key provisioning, cross-power replay resistance, or side-channel resistance.
- No full cryptographic proof; a 32-bit tag gives about 2^{-32} forgery probability.
- The serial link and CDC are the next stage; FPGA numbers await Quartus synthesis.

Appendix H. DE10-Nano Board Test

- Board: Terasic DE10-Nano, Cyclone V SoC (5CSEBA6U23I7), with Quartus Prime.
- Clock: `CLOCK_50` at 50 MHz directly, one clock domain.
- Procedure: synthesis (`quartus_sh --flow compile`), bitstream upload (`.sof/.rbf`), and real-time on-board testing.
- Frame loading: shift 192 bits (key 64, counter 32, payload 64, tag 32) over GPIO; the core loads the key and starts the MAC.
- SignalTap: `auth_ok`, `fresh_ok`, `done`, `host_full`, `fault`.
- On-board test: clean frame accepted (`host_full` high), corrupt frame rejected (`host_full` low, `fault` high), replay not committed.
- Reports: Fitter (ALM/FF/M10K/DSP), Timing Analyzer (Fmax, WNS), and PowerPlay.
- The organizer's FPGA/sandbox facility is used at the bootcamp.